Scheduler Equivalence and Performance Trade-offs under a Common LLM-Serving Contract

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Scheduler comparisons require more than identical inputs: the execution contract determines which policy distinctions can affect request outcomes. We examine four LLM-serving adaptations under a common contract. Under the original contract, fixed batch timing and a residency/work bound make chunked-FCFS and decode-first equivalent; complete terminal records confirm the equality in all 1,440 matched conditions, despite possible differences in journal ordering. A finite-work bound certifies urgency inactivity in both the original and budget-perturbed regimes; prompt-length admission ordering remains active. Among the three distinct behaviors, SLO-gated reduces aggregate mean latency by 11.3–34.0% against chunked-FCFS while reducing finite-window token throughput by 1.65–4.01%; BurstGPT mean window-p95 latency increases by 3.72% and 18.97%. Equal-window source means exhibit no strict-10% practical reversals in 2,376 comparisons, or 864 after removing duplicate behaviors and endpoints. Window-level and metric-dependent reversals persist: aggregate stability does not imply a common preference for every window or objective. A separate 72-cell budget sensitivity preserves the pair's observed equality on three input-selected BurstGPT windows, without proving equivalence under arbitrary binding budgets. These results connect effective scheduling behavior to the interpretation of its performance comparisons. The model isolates execution semantics; its rates do not predict a production GPU or establish a universal scheduler ranking.

CCS Concepts: • **Computing methodologies** → **Simulation evaluation**; • **Computer systems organization** → *Cloud computing*.

Additional Key Words and Phrases: LLM serving, performance evaluation, scheduling, trace-driven simulation, ranking stability

1 Introduction

Distinct scheduling priority lists can select the same work and induce the same request-state transitions. Aggregate rankings cannot distinguish this structural equivalence from coincidentally similar performance. Evaluation therefore requires establishing what the execution model allows each policy to change.

Workload sensitivity, metric dependence, and throughput–latency trade-offs are already well established [1, 4, 19]. Vidur and Vidur-Bench support shared simulation and workload/policy comparisons [3]; evaluation guidelines address baseline fairness and misleading summaries [2]. Those contributions motivate a more specific question: *which nominal scheduler distinctions survive a declared execution contract, why do others disappear, and what performance trade-offs remain?* Answering it requires examining state transitions and mechanism activation alongside performance measurements.

We investigate four project-maintained adaptations under CM-0.1, a contract specifying online information, legal actions, work advancement, batch timing, and resource acquisition and release. The adaptations retain selected ideas from FCFS serving, chunked prefill, decode-first scheduling, and SLAI-inspired prompt ordering [4, 6, 16]. Fixed batch cost removes composition-dependent service costs; four or eight resident slots preserve admission competition. This abstraction permits a precise account of executable distinctions, without predicting a particular GPU or comparing native serving systems. The complete selected corpus has 120 frozen production-derived windows, six joint arrival/deadline settings, and two residency capacities.

The scientific contribution connects three questions and their answers:

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RQ1: Effective behavior. Which policy distinctions change request execution? A residency/work bound and common batch-end timestamps imply identical per-request transitions for chunked-FCFS and decode-first under the original contract. All 1,440 matched terminal comparisons confirm this invariant; four labels yield three observed behaviors. A separate 72-cell budget sensitivity preserves the pair's equality in its selected stratum, an empirical finding beyond the original proof's conditions.
RQ2: Mechanism inactivity. Why does SLO-gated's urgency rule not execute? Clock updates give zero anchor age in the original contract. A finite-work bound also rules out first activation under the examined binding budgets. Completion deadlines do not enter the gate. The measured SLO-gated effects consequently cannot be attributed to urgency; prompt-length admission ordering remains an active distinction.
RQ3: Residual trade-offs and rankings. What preferences remain, and at what comparison level? Measured mean-latency improvements coexist with throughput losses and, on BurstGPT, larger mean window-p95 latency. Source-mean practical stability coexists with window-level and metric-dependent reversals. Removing duplicate behaviors and endpoints preserves that contrast, delimiting what an aggregate ranking can establish.

Section 3 gives the structural arguments; Sections 5 and 6 quantify the surviving comparisons. Event reconstruction and numerical verification support the findings. The separate historical diagnostic in Section 7 examines statistical calibration; it adds no common-contract observations. The selected windows support finite-corpus conclusions. Transfer to other execution contracts remains an open modeling question.

2 Related Work and Evaluation Scope

Simulation and benchmark comparisons. Vidur combines operator profiling and predictive modeling with hierarchical scheduling; its Vidur-Bench suite supplies workloads, scheduling/batching policies, and performance metrics, while Vidur-Search explores deployment configurations [3, Sections 4–6]. Beyond these established comparisons, we ask which policy distinctions the shared model preserves: Section 3 derives a conditional state-transition equivalence, and Section 3.4 bounds mechanism activation. LLMServingSim and its heterogeneous/disaggregated successor also model hardware/software interactions [9, 10]. Our fixed-duration work ledger instead permits direct reasoning about execution semantics. Journal reconstruction verifies conformance to that model; physical prediction would require separate validation [20].

Evaluation methodology and metrics. Papaioannou and Doudali study how datasets and inference scenarios change LLM-system evaluations [19]. Agrawal et al. identify anti-patterns in baseline comparisons, evaluation settings, and metric design, with a speculative-decoding case study [2, Sections 3–4]; Etalon introduces a fluidity index to capture streaming experience missed by conventional summaries [1, Sections 3–5]. We adopt their caution about benchmark and metric choices. Section 6 adds a matched account of source-mean, window-level, and cross-metric preferences after demonstrated redundancies are removed. Its absence of source-mean practical reversals has a specified threshold and denominator; it does not establish universal portability. We introduce neither a new streaming metric nor a general evaluation checklist.

Serving and scheduling mechanisms. Orca introduces iteration-level scheduling [26]; vLLM's PagedAttention addresses KV fragmentation and sharing [16]; Sarathi-Serve combines chunked prefill with stall-free batching [4]. Fu et al. learn relative output-length order to approximate shortest-job-first scheduling [12]. Our prompt-length ordering uses observed inputs, without predicting output length or claiming the same objective or optimality. Bari et al. give a formal

batching framework and distinguish the throughput-optimal RAD construction from practical SLAI, which combines decode urgency and known prompt lengths [6]. SOLA jointly considers request/system state and work allocation to improve SLO attainment [15]. These systems evaluate active state-dependent mechanisms under their own execution assumptions; our adaptation has a fixed, inactive urgency gate. Full-context reservations also omit native paging, while disaggregation and migration, exemplified by DistServe and Llumnix [22, 28], lie outside the evaluated contract.

Binding constraints and throughput. Dai et al.'s May 18, 2026 revision studies queueing stability with token-load- dependent batch time and a maximal-request-batch-size extension [11]. The binding constraint matters in both studies, but the estimands differ. Our contract fixes batch time, limits residency to four or eight requests, and measures output divided by finite-window completion duration. Its original-contract chunked work cannot exhaust the token budget; Section 5.6 separately tests lower budgets. These drained rates are not a stability region, and the observed throughput losses neither refute Dai et al.'s results nor establish a new optimality theorem.

Workload scope. BurstGPT and ServeGen characterize production-derived workloads [24, 25]; resampling has a longer history in scheduler evaluation [27]. Our analysis instead conditions on a fixed selected corpus, with equal-window source means and explicit matched comparisons. This supplies evidence within those windows, not a new workload generator or an estimate of production-wide scheduler preferences.

Inference and evidence reuse. STAT01 diagnoses established inference procedures [7, 8, 13, 14]; it introduces no new method. Historical inputs and rule outputs are reused, with separate common-model execution identities (Appendix B). Shared infrastructure does not constitute independent validation.

3 A Common Execution Contract

3.1 Why the comparison object matters

The historical corpus contains 9,360 configurations and two deterministic verification passes per configuration. Its primary panel contains eleven policies, with two additional style approximations. The historical engine permits policy-dependent service modes: instantaneous prefill for some policies, explicit prefill work for others, and differing enforcement of KV limits. A policy's requested work can also differ from the engine's realized progress. Consequently, those outcomes cannot be interpreted as a controlled comparison of scheduling order. Table 1 distinguishes the three evidence objects used in this paper.

The common contract distinguishes changes in the order of work that all executes from changes in which requests enter or progress. Fixing batch cost removes composition-dependent execution cost; bounded residency retains admission competition. We do not claim prospective selection of these settings for the equivalence or trade-offs reported here.

3.2 Service, resources, and information

CM-0.1 represents one abstract device. Each batch lasts 1 ms and has a 32,768-token work budget. Each executed prompt token consumes one unit; each selected decode request consumes one unit and emits one token. All policies share these rules. A request emits no output during its own prefill, including its final chunk; its decoding starts in a later batch. Other resident requests may decode in the same batch. Batch duration is independent of its composition. Thus seconds and tokens per second below are model quantities, not estimates for a particular GPU or LLM.

Each resident request reserves the complete 32,768-token context bound. All frozen request lengths fit; the maximum prompt-plus-output length is 30,053. Capacities of 131,072 and 262,144

Table 1. Evidence objects and the conclusions they support. Historical identifiers containing "faithful" are immutable names, not validation certificates.

Object	Comparison or check	Interpretation
Historical LSSP	Policy-dependent service/resource modes; archived ranking arithmetic	Motivation and historical-rule counts
C01-FULL	Four adaptations; common CM-0.1 execution and information	Descriptive finite-corpus performance
STAT01	Synthetic independent studies with known population truth	Scenario-specific inference calibration

Table 2. Project-maintained adaptations. Artifact IDs append `_c01` to `whole_fcfs`, `chunked_fcfs`, `decode_first`, and `slo_gated`. All use the shared contract above.

Adaptation	Historical lineage	Executed ordering rule and principal restriction
Whole-FCFS	vLLM-inspired whole-prompt implementation	Admit whole prompts in arrival order when feasible; otherwise decode residents. Admitting prefills defers all decodes that batch. No native paged allocator.
Chunked-FCFS	vLLM-inspired chunked implementation	Resident work before waiting work in stable order; at most 512 prefill tokens per request per batch.
Decode-first	Sarathi-inspired implementation	Resident decodes, then resident prefills, then waiting prefills; 512-token cap. No original engine timing.
SLO-gated	SLAI-inspired implementation	Urgent decodes first; then resident prefills, waiting prefills by prompt length, and residual decodes. Fixed 0.495 s gate; 512-token cap. No dynamic offset, heterogeneous classes, or full SLAI.

tokens therefore permit *four and eight resident requests*, respectively, even though the configured sequence limit is 64. Actual used KV follows executed prompt and output prefixes, rounded to 16-token blocks, and must remain within the reservation. Completion releases the reservation. Waiting requests own none. This conservative rule avoids output-length prediction and allocation deadlock; it does not model incremental native paging or cache eviction.

There is no preemption, dropping, migration, prefix sharing, or recomputation. Requests arriving during a batch become eligible at its end. Idle time advances to the next arrival. The administrative horizon is the final arrival plus 50 s; a batch cannot finish beyond it. Unfinished requests remain explicitly unfinished rather than contributing observed completion latencies.

Policies see only arrived request identifiers, stable order, arrival time, prompt length, acknowledged progress, residency, last-output/prefill-completion anchor, and current time. Actual remaining output length, future arrivals, source identity, and output-derived metric deadlines are hidden. The engine either executes the exact proposed token spans or rejects the plan. The shared contract therefore fixes both the resources and the online information available to each adaptation.

3.3 Adaptation identities and effective behavior (RQ1)

Table 2 defines the four evaluated adaptations. Published systems are lineage references; historical project implementations and CM-0.1 adaptations are separate objects. The panel was targeted using the historical comparison questions, not sampled to represent all schedulers.

Conditional state-transition equivalence. Let R be the resident count at a batch boundary and $K \in \{4, 8\}$ the reservation limit. The planner offers at most $K-R$ waiting requests a slot; a completing request releases its slot only after that batch's work. Each selected request appears once, contributing at most 512 prefill tokens or one decode token. Thus the bound is $512[R+(K-R)] \leq 4,096$, including both existing residents and new admissions, rather than a separate allowance for each. The 32,768-token budget cannot truncate this work.

Starting from the same state, chunked-FCFS and decode-first select the same residents and the same stable-order waiting prefix. They execute identical per-request spans, although the span lists can have different orders. The engine gives every span the same batch-end timestamp and releases completed reservations afterward. Consequently their next states, admissions, and timestamps agree inductively. The complete saved terminal records confirm this configuration-specific equivalence in all 1,440 matched conditions. The inductive argument establishes the invariant; agreement of terminal records checks its realization in the selected corpus. Equality of aggregate metrics alone would establish neither. This is not a theorem about the original serving systems.

The same work bound includes SLO-gated's residual decodes, even though they follow waiting prefills. A newly decode-ready resident's anchor is its prefill batch's end; every later decode resets it to the next batch end. There is no idle jump while such a resident exists. At each scheduling boundary its anchor age is therefore zero under the implemented clock updates, far below 0.495 s. The urgency gate never binds. Prompt-length ordering of waiting admissions remains active; resident prefills still progress, and later arrivals cannot preempt admitted requests. Whole-FCFS also retains a distinct rule: admitting whole prefills defers all decodes in that batch. Neither mechanism is removed by the work bound. Thus the model does more than give policies identical settings: it identifies which of their decision rules can change state transitions. The resulting finite-window completion distributions are the empirical object of Section 5.

3.4 Urgency inactivity and its sufficient condition (RQ2)

Urgency concerns a token gap, not a synthetic completion deadline. At a planning boundary t, an arrived, unfinished resident is critical exactly when its prefill is complete, its anchor a_i is defined, and $t - a_i \geq 0.495$ s. The anchor is the end of final prefill or the last executed decode; an omitted decode does not reset it. Waiting requests and unfinished prefills cannot qualify. Neither completion deadlines nor estimated remaining service enters this predicate. The protocol interprets 0.495 s as a uniform 0.5 s token-gap allowance minus a 5 ms margin, but the implementation fixes the threshold: it exposes no SLO parameter. Changing finite metric deadlines alone cannot change execution.

Finite-work inactivity certificate. A finite-work argument also limits activation when the budget binds. Let P be total prompt work, $B > K - 1$ the integer budget, K the residency limit, and δ the fixed batch duration. Suppose a first critical decision exists. In each consecutive batch omitting that ready decode since its anchor, the residual loop must have exhausted B; at most $K - 1$ other decodes consume one token each. Every such batch therefore executes at least $B - (K - 1)$ prefill tokens. With no recomputation or idle jump while the decode remains resident, m omitted batches imply

$$m[B - (K - 1)] \leq P, \qquad t - a_i \leq \delta \left\lfloor \frac{P}{B - (K - 1)} \right\rfloor. \tag{1}$$

Thus $\delta\lfloor P/[B - (K - 1)]\rfloor < 0.495$ s rules out first activation: the finite prompt work cannot sustain enough consecutive decode omissions to reach the urgency threshold. The common-domain restriction gives $P \leq NB$ for N requests. For $N = 200$, $B \geq 1,024$, $K \leq 8$ and $\delta = 1$ ms, the bound is at most 201 ms. This is a sufficient inactivity certificate for these assumptions, not a general

waiting-time guarantee or a necessary condition for activation. The original nonbinding case has the tighter zero-age proof above.

The fixed-threshold code agrees with its declared adaptation; configurable TBT classes in the historical scheduler are a different implementation. Engineering checks at the adjacent floating-point values below and above 0.495, and at equality, confirm the predicate without establishing that such supplied states are reachable. No new SLO/deadline experiment is run: the available metric deadline cannot exercise the gate, and altering the policy literal would change the frozen comparison object.

3.5 Conformance evidence

The event referee reconstructs request states, token spans, timing, completion, and resource use from the inputs and execution journal without importing the engine or policy implementation. It checks that intended and executed work agree and recomputes the metrics. Every campaign cell passed this check; an independent analyzer also checked the complete ranking and reversal reports. The final evidence review additionally checked all result identities and terminal metrics and replayed 24 structurally selected new-cell journals. These checks establish conformance to CM-0.1, not GPU fidelity. Figure 1 summarizes the evidence chain.

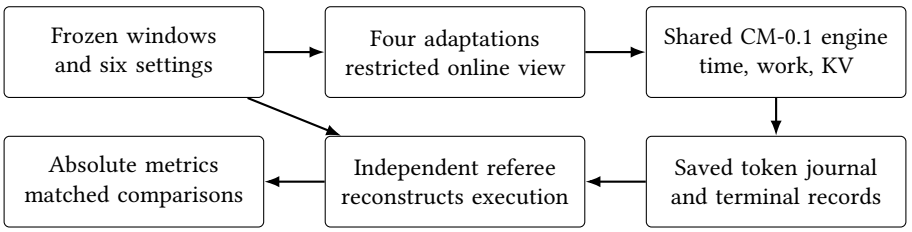


Fig. 1. Comparison and verification under one model. The independent analysis check recomputes aggregates, ranks, and reversals from the absolute metrics. No stage in this chain establishes physical-system accuracy.

4 Corpus, Conditions, and Analysis

4.1 Trace and implementation provenance

The corpus comprises 40 windows of 200 requests from each of Azure 2024, Bailian/Qwen, and BurstGPT [5, 18, 24]. The Azure release identifies DynamoLLM as its accompanying publication [21]; the pinned Bailian/Qwen release requests attribution to the KVCache workload study [23]. These are three separately released sources, not three independent provider populations: BurstGPT and Azure have Azure infrastructure in their collection provenance. Table 3 identifies the consumed files and transformations. The supplement records full file and window fingerprints. The official Azure conversation release has the name `AzureLLMInferenceTrace_conv_1week.csv`; its published SHA-256 matches the consumed file despite the local rename. Azure and BurstGPT are licensed CC BY 4.0, and the pinned Bailian/Qwen data Apache 2.0. Input acquisition instructions retain attribution, license texts, and transformation notices. Public artifact distribution remains pending.

Each source's 40-window panel contains ten inherited pilot windows and 30 subsequent windows selected using fixed seeds and valid-row ranges. The expanded manifest was frozen before its corresponding campaign outcomes; earlier pilot outcomes already existed. Within the C01 manifest, row-overlap checks find no overlapping selected ranges. Exact overlap with some related projects' older BurstGPT selections remains unresolved. Nonoverlap within this panel does not establish temporal independence or representativeness. The frozen chronology labels allocate 20/10/10

Table 3. Consumed dataset provenance. Each row contributes 40 windows and 8,000 request records. Hashes are prefixes of SHA-256 values; full values are in the bundled input manifest. The artifact provenance audit matches these hashes to upstream release assets or a pinned LFS object and retains the corresponding license notices.

Source	File and representation	File hash prefix
Azure 2024	`AzureLLMInferenceTrace_conv_2024.csv`; source arrival and token counts	`a0cc9b969a9b`
Bailian/Qwen	`qwen_traceB_blksz_16.jsonl`; input/output lengths read directly; 16-token prefix-block hashes are separate fields	`68e3f98e2d60`
BurstGPT	`BurstGPT_without_fails_2.csv`; retained valid source rows and token counts	`56193aa9b2bb`

windows per source to EARLY/MIDDLE/LATE. Four consecutive ten-window blocks provide an additional descriptive sensitivity; their elapsed time spans are unequal.

C01 reuses the historical input reconstruction, while its execution engine, policy views, and adaptation implementations have separate identities. C01-FULL retains C01 execution semantics at commit `9aed0206de66`; its full-campaign execution commit is `2d8342e699f3`. The complete matrix has $120 \times 6 \times 4 \times 2 = 5,760$ cells. Of these, 144 were reused from the pilot only after matching input, code, configuration, and journal identities; 5,616 were newly executed in the completed full campaign. Reuse is not an independent replication. No cell is omitted, and a deterministic cell is not rerun to manufacture a sampling distribution.

4.2 Inherited settings jointly transform arrivals and deadlines

The six labels LOW, PRE_KNEE, KNEE, POST_KNEE, OVERLOAD, and HIGH_PRESSURE come from historical calibration, not from a saturation analysis of CM-0.1. For a window-specific compression factor f, rebased arrival gaps and synthetic deadline slack are both divided by f. In idealized notation,

$$a'_i = (a_i - a_1)/f, \qquad d'_i = a'_i + (d_i - a_i)/f. \tag{2}$$

The implementation accumulates scaled nonnegative gaps in arrival order. Prompt/output lengths and priorities are unchanged. Factors range from 22.525 to 180.873 for Azure, 17.956 to 186.580 for Bailian, and 102.884 to 725.020 for BurstGPT. Equal labels therefore do not mean equal absolute arrival rates across windows or sources.

Deadlines are synthetic, derived in the inherited workload construction using output-length information; priorities are uniform. These attributes are not observed production service contracts. Deadlines are used only by the metric layer and are hidden from the scheduling adaptations. The setting axis is consequently *joint arrival/deadline stress*. A change in goodput across this axis cannot be attributed solely to arrival load.

4.3 Endpoints and aggregation

For arrival a_i, first output u_i, completion c_i, and output count o_i, latency is $c_i - a_i$ and time to first token (TTFT) is $u_i - a_i$. Time per output token (TPOT) is $(c_i - u_i)/(o_i - 1)$ for $o_i > 1$. The saved means and p95s use their defined completed-request populations; one-output requests do not have TPOT. The p95 uses the saved linear interpolation convention. Completion fraction is completed requests divided by all arrivals. Token throughput is completed output tokens divided by elapsed time from first arrival to final completion, or to the horizon if unfinished. Emitted-token throughput also counts any output of unfinished requests. Request throughput uses the same time

denominator. These are finite-window completion rates, including the drain period; they do not estimate a maximum sustainable arrival rate or system capacity.

Arrival-normalized weighted goodput is

$$\text{ANWG} = \frac{\sum_i w_i \, \mathbf{1}\{i \text{ completes and } c_i \le d_i'\}}{\sum_i w_i}. \tag{3}$$

Here $w_i = 1$, so ANWG is the fraction of arrivals meeting their synthetic deadline. It is dimensionless, unlike requests or tokens per second.

The eleven frozen ranking endpoints are ANWG, completion fraction, three throughput measures, and mean/p95 latency, TTFT, and TPOT. Larger is better for goodput, completion, and throughput; smaller is better for the six time endpoints. The complete endpoint set remains in the supplement, including redundant and numerically uninformative endpoints.

The window is the principal analysis unit. At a fixed source, setting, capacity, and policy, source means give each of the 40 windows equal weight. For the compact source–capacity table, we first average the six settings within each window and then average the 40 windows. A mean of window p95s is always labeled as such; it is not a pooled-request p95. Matched differences hold window, setting, and capacity fixed. We show all windows, not a selected favorable subset.

4.4 Ranking and practical-reversal definitions

Exact equality is primary: all maximizers or minimizers enter the top set, and tied policies receive average ranks. The frozen sensitivity admits values within $10^{-12}\max(1, |v_{\text{best}}|)$ of the best; it does not replace exact rankings or the practical-margin rule.

For policies A, B and condition c, let $\Delta_c = q_m(v_{A,c} - v_{B,c})$, where $q_m = 1$ for a larger-is-better endpoint and $q_m = -1$ otherwise. The relative margin is

$$r_c = \frac{|v_{A,c} - v_{B,c}|}{|v_{\text{loser},c}|}. \tag{4}$$

A practical reversal between c and c' requires $\Delta_c \Delta_{c'} < 0$ and both $r_c > 0.10$ and $r_{c'} > 0.10$. The strict boundary is evaluated exactly on the represented binary values. Zero losing values and missing values are unestimable. Nonzero ties remain eligible with margin zero. All denominators retain these distinctions.

Separate summaries compare within-source window pairs, Cartesian cross-source window pairs, settings within a window, capacities at a fixed window and setting, metrics at a fixed window, setting, and capacity, and source means at a fixed setting/capacity/metric. Cross-source windows are not paired by ordinal. For metric comparisons, each endpoint is oriented separately. These overlapping comparisons describe the finite corpus. We attach no population confidence intervals, p-values, or significance stars to C01.

5 Absolute Performance and Trade-offs (RQ3)

5.1 Completion and performance magnitudes

Every cell completes all 200 requests: 1,152,000 request executions across repeated conditions, not independent requests. There is no horizon censoring and no undefined cell-level latency, TTFT, or TPOT endpoint. TPOT populations range from 103 to 200 because single-output requests are excluded from that particular metric. Across cells, mean latency ranges from 0.008314 to 3.893923 s, p95 latency from 0.008560 to 7.300052 s, and token throughput from 332.489 to 7,766.685 tokens per simulated second.

Table 4. Absolute performance, equally averaging six settings within each window and 40 windows within each source. "Slots" are effective full-context residents. Mean and p95 columns are mean latency and *mean window-level p95 latency*; throughput is output tokens per simulated second.

Source	Slots	Adaptation	ANWG	Mean (s)	p95 (s)	Tokens/s
Azure 2024	4	Chunked / Decode	0.1746	2.8132	5.3232	3746.2
		Whole-FCFS	0.1746	2.8145	5.3275	3743.5
		SLO-gated	0.2001	1.8559	5.1342	3684.4
Azure 2024	8	Chunked / Decode	0.3174	1.3626	2.5986	7018.7
		Whole-FCFS	0.3101	1.4092	2.6892	6816.5
		SLO-gated	0.3892	0.9357	2.5131	6825.8
Bailian/Qwen	4	Chunked / Decode	0.1828	2.2655	4.2969	3603.2
		Whole-FCFS	0.1799	2.3020	4.3705	3548.3
		SLO-gated	0.1771	1.7205	3.9681	3531.4
Bailian/Qwen	8	Chunked / Decode	0.3330	1.0417	2.0227	6183.4
		Whole-FCFS	0.3189	1.1063	2.1463	5914.1
		SLO-gated	0.3693	0.8087	1.9037	5935.7
BurstGPT	4	Chunked / Decode	0.1030	0.5225	1.0005	3354.9
		Whole-FCFS	0.0889	0.5654	1.0843	3054.9
		SLO-gated	0.2321	0.4448	1.0377	3223.4
BurstGPT	8	Chunked / Decode	0.3072	0.1907	0.3707	5532.1
		Whole-FCFS	0.2129	0.2314	0.4505	4796.5
		SLO-gated	0.4613	0.1692	0.4410	5322.3

Table 4 reports absolute source–capacity aggregates. Chunked-FCFS and decode-first share a row because their terminal records and all metrics are exactly equal; their separate identities remain in the artifact.

5.2 Mean latency gains do not imply a universal improvement

SLO-gated reduces aggregate mean latency against chunked-FCFS in all six source–capacity groups, by 11.3–34.0%. The largest aggregate reduction is Azure at four slots, from 2.8132 to 1.8559 s. At eight slots, BurstGPT changes from 0.1907 to 0.1692 s, the smallest percentage improvement. This is a substantial difference in the model's mean response time, but it comes with a throughput reduction in every group, ranging from 1.65% to 4.01%.

The tail comparison is less favorable. For BurstGPT, mean window-p95 latency rises from 1.0005 to 1.0377 s at four slots and from 0.3707 to 0.4410 s at eight slots: increases of 3.72% and 18.97%. Shorter average latency can therefore coexist with longer upper-tail latency under the same contract. At four-slot Bailian, ANWG is also 3.12% lower for SLO-gated despite its 24.05% mean-latency reduction. Throughput, deadline attainment, and response time reward different aspects of the completed schedule.

For a matched, fully completed 200-request condition, mean latency is $\sum_i (c_i - a_i)/200$, whereas token throughput is $\sum_i o_i/(\max_i c_i - \min_i a_i)$. Reducing the sum need not reduce the last completion or an upper quantile. Deadline attainment counts which completions cross synthetic deadlines; its Bailian loss is therefore compatible with the mean-latency gain. The source/capacity throughput means average window rates and are not inverses of mean durations. The following saved-record illustration connects these distinct objectives to the active admission ordering identified in Section 3.

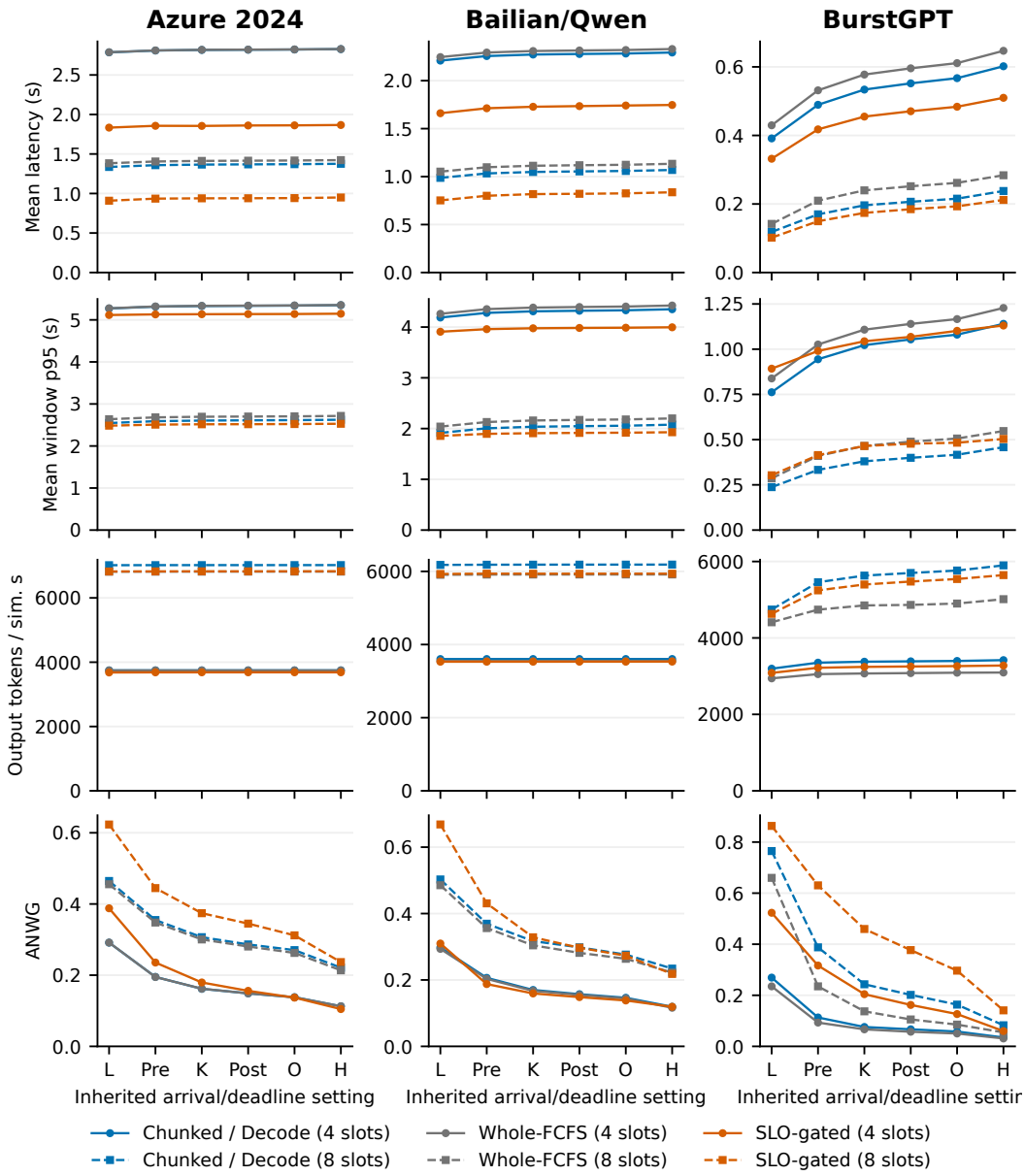


Fig. 2. Absolute performance across all six inherited settings, equally averaged over 40 windows per source. Solid circles: four slots; dashed squares: eight slots. L/Pre/K/Post/O/H abbreviate LOW through HIGH_PRESSURE in their declared order. Chunked and decode-first curves coincide exactly. The p95 panels average window-level p95s. Both arrivals and synthetic deadlines change along the horizontal axis. Lines connect descriptive condition means, with no inferential uncertainty bands.

5.3 A saved-record explanation of the trade-off

Consider BurstGPT window `burstgpt_pilot_v2_w12`, setting `HIGH_PRESSURE`, capacity 262,144 (eight resident slots). We selected this illustration retrospectively: among BurstGPT conditions where SLO-gated has lower mean latency, lower throughput and higher p95 than chunked-FCFS, it is first in lexical window/setting/capacity order. This is an outcome-selected example, not a random or representative window. Its identity is frozen in `major-revision/example-manifest.json`.

Both adaptations complete the same 200 requests and emit 3,520 output tokens. Table 5 uses all requests: mean latency falls by 40.60%, but p95 rises by 33.17% and throughput falls by 14.63%. The throughput denominator runs from first arrival to final completion, including drain; it grows from 1.150 to 1.347 s. The p95 linearly interpolates ordered per-request latencies at zero-based position 189.05. It is a single-condition quantile, not the source mean of window p95s in Table 4.

Table 5. One outcome-selected BurstGPT condition, with identical completed inputs and output count. All times and rates are CM quantities. Duration includes the drain; percentages in the text are relative to chunked-FCFS.

Adaptation	Mean (ms)	p95 (ms)	Duration (s)	Tokens/s
Chunked-FCFS	61.681	110.741	1.150	3060.870
SLO-gated	36.641	147.480	1.347	2613.215

The saved journals show how admission competition remains active despite the nonbinding token budget. At 14 ms, request 5 has just completed, leaving one slot; requests 13 and 14 have both arrived. Chunked-FCFS admits request 13 (214 prompt tokens), whereas SLO-gated admits request 14 (201 tokens). The other request enters at 16 ms in each run. This is the first difference in admitted membership, not merely in the order of spans within a batch. Table 6 includes this pair, the request with the largest completion delay (26), and the request with the largest improvement (196). Waiting here means arrival to the start of first prefill, not TTFT or total latency; completion timestamps are measured from the common origin.

Table 6. Selected request timelines from the same journals. C: chunked-FCFS; S: SLO-gated. Request IDs are zero-based saved input IDs. Output lengths are used retrospectively for explanation and were hidden from both policies.

ID	Prompt/output (tokens)	Arrival (ms)	Wait to prefill		Completion time	
			C (ms)	S (ms)	C (ms)	S (ms)
13	214/7	13.399	0.601	2.601	22.000	24.000
14	201/7	13.399	2.601	0.601	24.000	22.000
26	1976/1120	24.883	1.117	198.117	1150.000	1347.000
196	189/7	178.009	104.991	1.991	291.000	188.000

Request 26 has a 1,976-token prompt and 1,120 output tokens. Its first prefill starts at 26 ms under chunked-FCFS and 223 ms under SLO-gated; its first output occurs at 31 and 228 ms, respectively. It completes last in both runs, at 1.150 and 1.347 s. Thus its later service accounts exactly for the longer denominator in this condition, without changing total output. The admission rule observes its prompt length, not its output demand. Across all requests, 151 complete earlier under SLO-gated, 29 later and 20 at the same time; the sum of latency changes is −5.008 s. Mean latency therefore improves even while the last completion and upper quantile worsen. The p95 brackets involve

other requests (191/195 under C and 139/68 under S), so the final request alone does not explain the p95 change. These timelines explain compatibility of the objectives in a real saved condition; the few displayed requests do not exhaustively decompose the aggregate change or estimate causal mechanism effects across the corpus.

The example manifest records exact input/result/journal identities and SHA-256 hashes. Independent event reconstruction checks all 400 terminal states and 2,497 events against the displayed tables; the complete records support reproduction of the illustration without selecting a new condition.

5.4 Capacity and inherited-setting effects

Increasing reservation capacity doubles permitted residency from four to eight. In every source–adaptation aggregate shown, mean and mean-p95 latency fall and throughput rises. For chunked-FCFS, Azure mean latency falls from 2.8132 to 1.3626 s while throughput rises from 3,746.2 to 7,018.7 tokens/s; BurstGPT changes from 0.5225 to 0.1907 s and from 3,354.9 to 5,532.1 tokens/s. The capacity axis is predominantly a concurrency constraint in this model. The largest observed used KV is 47,872 tokens in the four-slot profile and 62,880 in the eight-slot profile, both below even the smaller nominal capacity. These results do not measure native allocator pressure, eviction, or the benefit of paging at an exhausted used-token limit.

Figure 2 separates settings within each source–capacity. The inherited setting labels can produce large changes in goodput with small changes in latency or throughput. For Azure chunked-FCFS at four slots, LOW to HIGH_PRESSURE changes mean latency from 2.7858 to 2.8269 s and throughput from 3,745.9 to 3,746.3 tokens/s, while ANWG falls from 0.291375 to 0.112625. Since the transformation also contracts deadline slack, this is not evidence for an isolated arrival-load effect on goodput.

5.5 Matched-window variation

Figure 3 displays all 40 window-level matched differences per source and capacity, averaging the six paired setting differences within each window. It complements the source means by showing where a performance difference changes sign or is concentrated. The vertical zero line represents equal performance; it is neither a confidence bound nor the 10% practical-reversal threshold. For latency, a negative difference favors SLO-gated; for throughput, it favors chunked-FCFS.

At the unaveraged condition level, SLO-gated uniquely minimizes mean latency in all 480 Azure conditions, 468 of 480 Bailian conditions, and only 198 of 480 BurstGPT conditions. Its favorable source aggregates therefore do not mean that it wins every window/setting comparison. The tail and throughput distributions make the same distinction visible without reducing each comparison to a winner label.

Performance component of RQ3. The surviving behaviors exchange average response time against throughput, tail latency, and deadline attainment. The capacity contrast is principally a residency constraint; the setting contrast jointly stresses arrivals and deadlines. These are trade-offs within the declared model.

5.6 A prospective token-budget sensitivity

To test a boundary of the equivalence argument, we prospectively specified that reducing only the batch budget would make chunked-FCFS and decode-first execute different per-request spans in at least one matched condition. The predicted mechanism was prefill work exhausting the budget before later resident decodes. The unmodified engine requires every whole prompt to fit the budget for all policies. Full-corpus feasibility requires at least 29,067 tokens, above the 4,096-token chunked-work bound. No intact Azure or Bailian window fits below 4,096 tokens; using them would

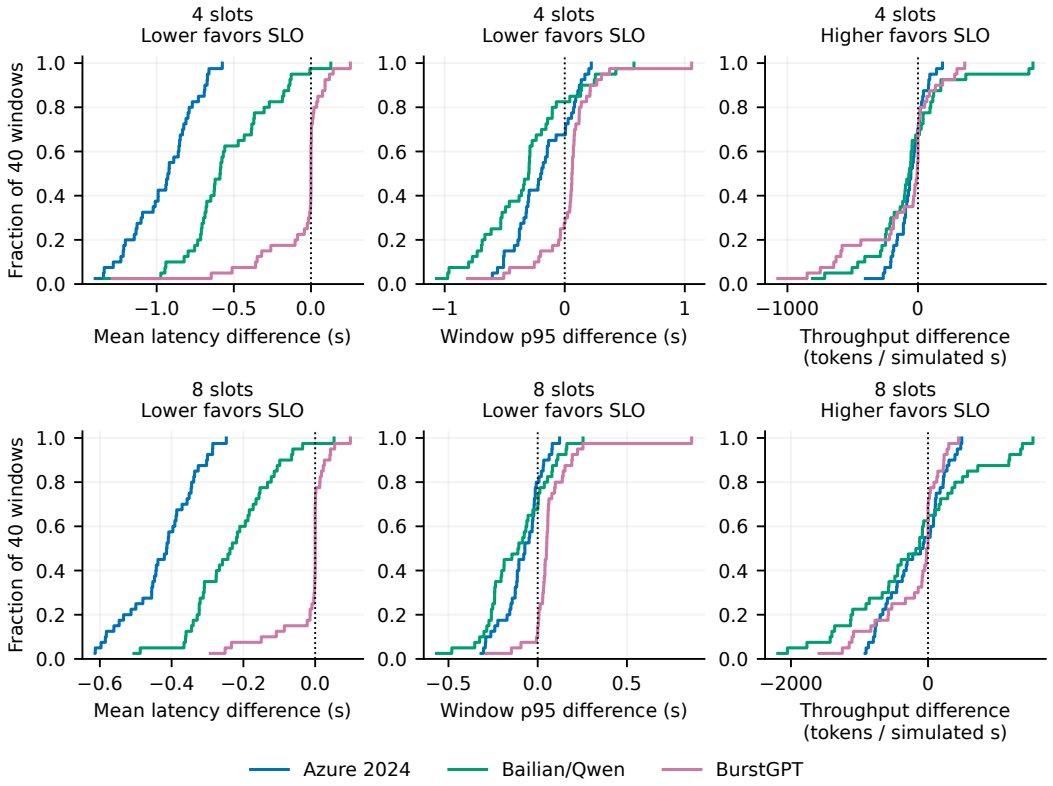


Fig. 3. Empirical distributions of SLO-gated minus chunked-FCFS differences. Each curve has 40 equally weighted windows; each point averages six matched setting differences within one window. The axes show absolute differences, not percentages. Positive latency differences and negative throughput differences are unfavorable to SLO-gated. All windows are retained.

require another contract or input change. We therefore selected the lexicographically first three frozen BurstGPT windows with maximum prompts strictly above 512 and at most 1,024 tokens: `pilot_v2_w16`, `w23`, and `w28`, with maxima 1,005, 790, and 648. This input-selected stratum permits multi-batch prefills but is not representative of the full corpus.

All 200 requests, frozen seeds and arrival/deadline transformations remain unchanged. We retain LOW and HIGH_PRESSURE, eight resident slots, and all four adaptations. Budgets are the original 32,768, then 2,048 and 1,024 tokens, corresponding to four and two 512-token chunks versus the 4,096-token potential work bound. Timing, reservations, horizon and policy code are fixed. A 12-cell feasibility pilot precedes the remaining 60 cells; all 72 pass the independent event referee, complete all 14,400 request executions, and have independently recomputed terminal metrics. The 24 baseline cells exactly reproduce their archived metrics and terminal states.

The primary hypothesis is rejected for this design: normalized token-span journals and terminal states agree for chunked-FCFS and decode-first in all 12 perturbed matches, and three distinct terminal behaviors remain in each of the 18 budget–condition combinations. Both policies serve every ready resident decode in these runs; budget exhaustion changes prefill progress without separating their transitions. This observation extends the tested operating region, not the nonbinding-bound

proof to arbitrary budgets. Span-list ordering differs in four of six matches at every budget, but causes no state or timing difference.

The perturbation does affect execution. Across all four labels, exactly full-budget batches increase from 0/25,983 to 3/25,986 and 36/25,984 as the budget decreases. These counts include the equivalent labels and exclude whole-prompt packing that leaves unusable residual budget. At 1,024 tokens, SLO-gated's prefill-before-residual-decode rule omits 42 ready-decode opportunities out of 35,976; its maximum anchor age reaches 2 ms, still below the unchanged 495 ms urgency threshold. The gate never activates. Equation 1 strengthens this observation: total prompt work of 39,300, 52,250 and 44,765 tokens gives respective 38, 51 and 44 ms upper bounds at budget 1,024, ruling out first activation in each window. Independent reconstruction of all 18 saved SLO-gated cells finds zero urgent decisions in 107,844 ready-decode opportunities. Table 7 shows modest, sometimes nonmonotone terminal effects. Whole-FCFS changes which whole prompts fit and hence which batches defer decodes; a smaller budget need not worsen every completion time.

Table 7. Budget sensitivity: minimum and maximum paired changes from 32,768 tokens across six fixed window–setting conditions. Latency columns are milliseconds; throughput is model tokens/s. Decode-first equals chunked-FCFS in every row. Ranges describe these conditions, not confidence intervals. All other contract parameters remain fixed.

Budget	Adaptation	Δmean	Δp95	Δthroughput
2048	Whole-FCFS	[0.000, 0.565]	[0.000, 1.000]	[-10.931, 0.000]
2048	Chunked-FCFS	[0.000, 0.135]	[0.000, 0.000]	[0.000, 0.000]
2048	SLO-gated	[0.000, 0.015]	[0.000, 0.000]	[0.000, 0.000]
1024	Whole-FCFS	[-2.245, 2.560]	[-7.000, 4.000]	[-18.604, 18.332]
1024	Chunked-FCFS	[0.000, 0.520]	[0.000, 1.897]	[0.000, 0.000]
1024	SLO-gated	[0.000, 0.365]	[0.000, 2.000]	[-4.503, 0.000]

No strict-10% policy-pair preference reverses across budgets (0/576 eligible comparisons: six conditions, two perturbations, six policy pairs, eight endpoints). With the existing numerical-tie tolerance, 5/96 top sets change, all involving TPOT as SLO-gated leaves a former tie; its largest mean-TPOT increase is only 5.87 μs. Whole-FCFS deadline attainment changes by −0.005 to +0.010, and SLO-gated by −0.005 to zero. All throughput changes are below 0.4% in magnitude. These bounded findings preserve the tested equivalence while exposing small execution changes; they do not retest cross-source means or establish equivalence outside this stratum. Three selected, temporally related windows support paired descriptive ranges, not population confidence intervals. The original C01 and reduced-comparison results remain separate and unchanged.

6 Rankings, Aggregation, and Metric Choice (RQ3)

6.1 Stable practical source comparisons and heterogeneous windows

Table 8 reports the strict-10% reversal rule with eligible and total denominators. Across all source-mean comparisons the count is **zero out of 2,376**. Each source mean uses all 40 windows at a fixed setting, capacity, and metric. The denominator is 3 source pairs ×6 settings ×2 capacities ×11 endpoints ×6 policy pairs. For this equal-window estimand, changing source does not overturn a policy-pair preference by more than the declared margin in both directions. The 2,376 comparisons include the equivalent chunked/decode-first pair and redundant throughput endpoints; they do not constitute 2,376 independent confirmations or establish population equivalence.

This null result is compatible with smaller sign changes and different tie sets. Among 396 comparisons of source pairs at fixed settings, capacities, and metrics, exact top sets are equal in 307

Table 8. Strict-10% practical reversals. Nonzero ties remain in eligible denominators; a zero losing value is unestimable. All counts include six policy pairs and the declared endpoints. Reused windows and overlapping pairs make these dependent descriptive counts, not independent trials.

Comparison axis	Reversals	Eligible	Total	Unestimable
Within-source windows	34,394	1,847,700	1,853,280	5,580
Cross-source windows	109,913	3,788,640	3,801,600	12,960
Inherited settings	920	236,853	237,600	747
KV capacities	350	47,418	47,520	102
Metric pairs	11,795	473,580	475,200	1,620
Source means	0	2,376	2,376	0

Table 9. Ranking summaries, each out of 1,440 window/setting/capacity conditions. "Near-tie change" counts conditions whose top set changes under the predeclared $10^{-12}\max(1, |v_{\text{best}}|)$ tolerance. No ranking is missing. "SLO unique top" always uses exact equality.

Endpoint	Exact top tie	SLO unique top	Near-tie change
ANWG	425	1,004	0
Mean latency	289	1,146	6
p95 latency	586	834	0
Token throughput	906	482	0
Completion	1,440	0	0
Mean TPOT	979	461	1,129
p95 TPOT	1,348	92	104

and intersect in 323; complete average-rank vectors are equal in 247. Absence of a strict practical reversal is weaker than equality of rankings or top sets.

Window-level variation remains visible. Within-source window comparisons yield 34,394 reversals among 1,847,700 eligible comparisons; Cartesian cross-source window comparisons yield 109,913 among 3,788,640. For ANWG alone, the corresponding counts are 8,472/162,900 and 35,352/332,640. The larger window-pair denominators arise from comparing every unordered pair of 40 windows within a source (780) or every Cartesian pair across sources (1,600). They are not larger independent sample sizes. These summaries cannot establish that one axis causally contributes more instability than another.

Within the same window, settings produce 920/236,853 eligible reversals, capacity changes 350/47,418, and metric choice 11,795/473,580. The last count compares policy-pair directions across endpoints; it demonstrates that a preference can depend on the objective even with the workload and resources fixed. It is distinct from comparing top sets, which asks whether at least one best policy is shared rather than whether a particular pair reverses.

6.2 Ties and redundant endpoints

Table 9 retains exact ties instead of breaking them by policy identifier. SLO-gated is uniquely best for mean latency in 1,146 conditions and for throughput in 482. Completion has a four-way tie everywhere.

All requests finish, making emitted-token and completed-token throughput identical. Request throughput differs only by a workload-specific multiplier shared by all policies at a fixed condition. The three throughput rankings therefore coincide. Chunked-FCFS, decode-first, and SLO-gated

Table 10. Post hoc reduced view: three behavior pairs and eight endpoints. The original strict-10% rule is unchanged. These dependent descriptive counts supplement, rather than replace, Table 8.

Axis	Reversals	Eligible	Total
Source means	0	864	864
Within-source windows	18,732	670,637	673,920
Cross-source windows	61,880	1,374,720	1,382,400
Settings	629	85,962	86,400
Capacity	240	17,219	17,280
Metric	4,492	120,288	120,960

also produce essentially one output per millisecond after decoding begins: the largest chunked-versus-SLO mean-TPOT difference is 1.80×10^{-16} s. Timestamp subtraction nevertheless changes exact top sets; the predeclared near-tie tolerance changes mean-TPOT top sets in 1,129 conditions and p95-TPOT top sets in 104. These tiny distinctions have no practical performance meaning.

Across all 79,200 metric-pair ranking comparisons, exact top sets are equal in 27,915 and intersect in 57,602. These counts include redundant throughput endpoints, completion ties, and numerical TPOT distinctions. They summarize reporting consequences; they are not eleven independent tests of a scheduler's robustness.

6.3 Post hoc view of distinct behaviors and metric families

To test whether duplication drives the qualitative ranking conclusions, we project the saved outputs onto whole-FCFS, chunked-FCFS, and SLO-gated. We remove the universally tied completion endpoint and retain token throughput as the representative of the three identical throughput rankings. The resulting eight endpoints cover five families: latency, TTFT, TPOT, throughput, and deadline attainment. Both mean and p95 remain for the three time families because they describe different distributional features. TPOT remains informative for whole-FCFS: its mean and p95 exceed the best value by the same strict-10% practical margin in 341 and 1,027 conditions, respectively. Removing TPOT altogether would discard that contrast.

This view contains 11,520 rankings, 34,560 matched pairwise records, and 8,208 reversal groups. Independent exact-boundary calculations from absolute metrics reproduce every retained archived group. Table 10 still has no source-mean practical reversals (0/864: three source pairs, six settings, two capacities, eight endpoints, and three behavior pairs), but has window and metric reversals. This is stability of equal-window source means under the strict-10% rule, not universal ranking portability. The qualitative contrast therefore survives removal of these demonstrated redundancies. Rates change with the denominator; the remaining endpoints and overlapping comparisons are not independent replications. The original four-label, eleven-endpoint outputs remain intact. Chronological sensitivities appear in Appendix C.

Ranking component of RQ3. Source-mean practical stability does not imply uniform preferences across windows or objectives. This remains true after collapsing duplicate labels and endpoints; magnitudes and top sets distinguish meaningful preferences from ties and numerical noise.

7 Supporting Historical Interpretation and Calibration

7.1 What the historical comparison establishes

The historical analysis reported 36 of 990 policy-pair/source-pair/setting records meeting its practical-margin and nominal-interval rule. Numerical reproduction confirms that archived-rule

Table 11. Compact calibration diagnostic for the actual historical-source screened-BH comparator. Each row has 8,000 studies; ordinary fixed-contrast interval coverage is available in all of them. Bounds are the expanded 17-method simultaneous Monte Carlo bounds, not confidence intervals for C01 performance. These three null scenarios isolate IID, independent-block, and AR dependence; all 18 scenarios and methods remain in the supplement.

Scenario	FDR	MC bounds	Coverage	MC bounds
IID equal null	0.09025	[0.06878, 0.11172]	0.94187	[0.92357, 0.96018]
Independent-block null	0.91800	[0.89726, 0.93874]	0.64762	[0.61465, 0.68060]
AR(0.7) null	0.98037	[0.96780, 0.99295]	0.56487	[0.53081, 0.59894]

count. It does not establish population reversals: the historical policy–model combinations differ, selected trace windows have no known population sampling law, and the inference procedure's calibration was unestablished. The C01 panel and execution model change the comparison object, so its zero source-mean practical reversals are not a like-for-like disappearance of those 36 records. Historical and common-model outcomes are kept in separate evidence strata rather than combined into a leaderboard.

7.2 A focused known-truth diagnostic

STAT01 evaluates historical inference in synthetic studies whose population means, covariance, and missingness are specified. Each study has three sources, eleven policies, and normally 40 windows per source; one bounded scenario uses 400. The 165 source-pair/policy-pair hypotheses are dependent within a study. Independently generated studies, not their individual hypotheses, are the Monte Carlo units. The completed design contains 134,000 studies across 18 scenarios, including IID, block-dependent, autoregressive, nonnormal, bounded, and missing-data cases.

The primary truth is a population reversal exceeding 10% relative to the losing mean in both sources. Sign reversals are a separate endpoint. The historical procedure first screens noisy point estimates at 10%, then uses bootstrap intervals and zero-effect p-values on the selected contrasts. Testing a zero-effect null after such screening is not automatically a test of the practical-effect null. BH or BY alone cannot supply the missing calibration [7, 8].

Three implementation identities are retained. The eight frozen original procedures include a port of the historical screened-BH procedure. A post-diagnosis comparator reproduces the actual historical source's floating-point evaluation order. Eight additional comparisons implement explicit mathematical threshold semantics. Thus the expanded family has 17 methods, with nine additions identified as post-diagnosis. Arithmetic correction is not a new test and does not change the inferential assumptions. Appendix A gives the definitions and the saved boundary case.

Table 11 shows that the actual historical-source procedure has practical FDR 0.09025 in the IID equal-null scenario, with an expanded Monte Carlo lower bound above 0.05. Independent-block and AR nulls show much larger false-discovery rates and poor ordinary interval coverage. These null examples already demonstrate a calibration concern without requiring interpretation of power or near-boundary alternatives.

Appendix A retains all classifications, stopping counts, original-eight and expanded-17 uncertainty families, abstentions, precision gaps, and low-power outcomes. These qualify the calibration findings.

Implication for the common-contract study. Reproducing historical arithmetic establishes what its rule reported; execution conformance establishes that C01 follows its common model. Neither

establishes valid statistical inference. STAT01 demonstrates calibration failures for historical procedures in explicit known-truth scenarios, while retaining uncertainty and assumption-specific limits. C01's finite-corpus descriptions do not depend on population significance tests.

8 Discussion and Threats to Validity

From policy labels to interpretable comparisons. The structural analysis identifies which decision rules can change execution before interpreting their rankings. Nonbinding work and common batch-end updates erase the chunked/decode-first priority distinction; admission competition and whole-prefill/decode exclusion survive. The urgency certificate supplies a different kind of result: a bound on reachable anchor age rules out an otherwise implemented branch. Together these explain why a nominally richer policy panel need not represent more executed mechanisms.

The residual mean/tail/throughput trade-offs are established scheduling phenomena, measured here within that reduced behavior set. The saved BurstGPT timeline illustrates how many earlier completions can lower mean latency while a later final completion extends the drain. It does not establish optimality or a corpus-wide causal decomposition. Source means answer an equal-window average question; window reversals expose variation concealed by that average, while cross-metric reversals concern different objectives on the same workload. Their coexistence, even in the reduced analysis, shows why stable aggregate preferences cannot select a scheduler for every window and objective. Exact top sets further distinguish ties from strict preferences.

Model adequacy and transfer. Fixed batch timing and conservative full-context reservations permit the state-transition argument but omit composition-dependent efficiency, kernel, bandwidth, communication, and native allocation costs. A binding work budget can invalidate the proof's premise without breaking equivalence: the focused sensitivity preserves terminal equality while changing some admissions and decode service. Budget saturation is rare in its short-prompt BurstGPT stratum. The finite-work certificate rules out urgency activation there; completion deadlines remain hidden from the gate. Longer prompts fail the unchanged common-domain check at these budgets, and composition-dependent timing remains untested. These restrictions are central to the explanation and limit its transfer to hardware-aware models and serving systems. Conformance checks establish implementation correctness for the abstraction, not physical prediction accuracy [20].

Corpus and inference limits. The selected windows are finite and temporally related, and the Azure-linked sources are not independent provider populations. Nonoverlap inside the C01 panel does not settle overlap with older cross-project BurstGPT selections. Uniform priorities and synthetic output-derived deadlines restrict goodput interpretation; the inherited transformation changes deadlines and arrivals together. C01 consequently reports descriptive finite-corpus effects. The reduced analysis is post hoc and removes only demonstrated redundancies; its remaining metric families and overlapping comparisons are still dependent.

STAT01's known truth applies only to its generators. Post-diagnosis methods, finite study counts, conditional interval availability, and conservative power remain explicit in Appendix A. Reproducible arithmetic and valid population inference are distinct requirements. The extensive checks reduce implementation and reporting risk without adding scientific novelty or resolving model adequacy by themselves.

9 Conclusion

Under the original common contract, a work bound and shared clock updates make two nominal adaptations state-equivalent. A separate finite-work certificate explains why decode urgency cannot activate in the examined regimes. The surviving behaviors retain mean/tail/throughput trade-offs,

and their preferences depend on the comparison level: source-mean practical stability coexists with local and cross-metric reversals after demonstrated redundancies are removed. The budget sensitivity preserves the observed equivalence in a small selected stratum, without proving it for arbitrary binding budgets. The contribution connects model constraints, effective behavior, and the interpretation of performance comparisons.

Code and Data Availability

The manuscript's reproduction materials include source, figure and table scripts, frozen analysis summaries, input fingerprints, and supplementary CSVs containing all 5,760 C01 cells and 7,200 STAT01 endpoint summaries. A separately assembled local artifact contains the pinned execution code and complete saved campaign records. Regenerating displays from those records is distinct from acquiring upstream inputs and replicating the experiments. Versioned public distribution of the complete research artifact is pending; no archival DOI is asserted. Original third-party raw workload traces are excluded from the planned release and must be acquired from the attributed upstream sources. The intended license scopes are MIT for original code and CC BY 4.0 for eligible original research data and results, with upstream terms retained.

Declarations

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AI assistance. OpenAI Codex was used to assist with software development, experimental analysis and validation, preparation of figures and tables, and manuscript revisions. OpenAI ChatGPT was used to assist with scientific brainstorming, manuscript organization, and suggestions for presenting and explaining the research. The author reviewed the resulting work and accepts full responsibility for the manuscript's content, scientific accuracy, and conclusions.

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A Statistical Diagnostic: Definitions and Complete Scenario Summary

A.1 Study construction and truth

A study draws a latent array $Y \in \mathbb{R}^{3\times n\times 11}$, with positive population means μ_{sj}. Sources are independent within this synthetic design. Dependence across policies is $C(\rho) = (1-\rho)I + \rho\mathbf{1}\mathbf{1}^{\mathsf{T}}$, usually with $\rho = 0.5$. Temporal covariance is identity, independent five-window blocks with within-block correlation 0.7, or AR(1) with correlation $0.7^{|i-j|}$. Source-specific innovations have mean zero and variance one; a Cholesky transformation imposes the specified covariance, with default scale $\sigma = 0.15$. The exact-ties scenario uses $\rho = 1$.

Each family comprises $\binom{3}{2}\binom{11}{2} = 165$ hypotheses. For a source pair (s,t) and policy pair (a,b), sign-reversal truth is $(\mu_{sa} - \mu_{sb})(\mu_{ta} - \mu_{tb}) < 0$. Practical-reversal truth is

$$\begin{aligned}&(\mu_{sa} > 1.1\mu_{sb} \ \wedge \ \mu_{tb} > 1.1\mu_{ta})\\ &\quad \vee \ (\mu_{sb} > 1.1\mu_{sa} \ \wedge \ \mu_{ta} > 1.1\mu_{tb}).\end{aligned} \tag{5}$$

Equality at 10% is null. Truth is enumerated from decimal/rational mean specifications, independently of estimated means and test decisions.

Equal-null means are 0.5. Stable and near-tie means are $0.5 + 0.01j$ and $0.5 + 0.0002j$, respectively. A reversal of size δ assigns source 0 policies 0/1 means $0.5 \pm \delta/2$, swaps them in source 1, and sets all other means to 0.5. Small, medium, and large IID reversal scenarios use $\delta = 0.03, 0.10, 0.30$. The boundary scenario uses 0.55/0.50 and the reverse. Block, AR, unequal-variance, mixed-distribution, and MCAR reversal scenarios use $\delta = 0.15$. Unequal scales are $(0.08, 0.25, 0.15)$; mixed innovations are standardized normal, Student-t_5, and lognormal. MCAR retains each policy/window observation independently with probability 0.8. Selective missingness removes source 0/policy 0 and source 1/policy 1 observations below their respective latent means; truth remains the unconditional mean. Observed values cannot identify that target merely by ignoring missingness.

The bounded scenarios use IID standardized uniform innovations, $\sigma = 0.08$, and $\delta = 0.30$, giving scores in $[0,1]$ analytically, without clipping. All scenarios have 40 windows per source except the last, which has 400. Raw arrays and masks are retained in the full local artifact candidate. The 112,275 reused and 21,725 subsequently generated studies sum to the completed 134,000-study design; no study was removed because of a numerical disagreement between implementations.

A.2 Frozen methods and post-diagnosis additions

All methods use nominal level $\alpha = q = 0.05$. The historical screened procedure uses 2,000 bootstrap draws of matched observed per-window policy differences, percentile intervals, and $p = \min\{1, 2\min[\Pr_{\text{boot}}(\bar{D} \le 0), \Pr_{\text{boot}}(\bar{D} \ge 0)]\}$. It combines source-side p-values by their maximum, applies BH to the point-screened family, and requires both intervals to exclude zero. The full-family variant assigns untested hypotheses p-value one and applies BH over all 165. The historical plus-one BY variant transforms the component bootstrap p-value as $\min\{1, (2000p + 2)/2001\}$

before source combination and applies BY over 165. These modifications do not make selection or the original p-values valid by definition.

The analytic practical-effect methods use paired contrasts $Z = Y_a - 1.1Y_b$ and their reversed orientation. Each orientation combines the two one-sided source tests by a maximum (an intersection–union test). The union over orientations uses $\min\{1, 2\min(p_1, p_2)\}$ before Holm correction over 165 hypotheses [14]. IID t-Holm uses paired Student tests; cluster t-Holm uses independent five-window cluster averages, retaining only completely observed clusters. Fewer than two complete clusters causes abstention. Their finite-sample Gaussian justifications apply to their respective independent sampling units with complete or MCAR observations, not arbitrary temporal dependence. The covariance-oracle normal variant uses the known Gaussian covariance of each observed matched sample mean. It does not access missing latent outcomes and is not deployable without that covariance information.

Grouped-bootstrap BY resamples nonoverlapping five-window groups jointly across policies, preserving masks and pairing. Its centered one-sided component p-value is $(1+\#\{\bar{Z}^* - \bar{Z} \geq \bar{Z}\})/2001$. A zero-observation resample causes component abstention; it is not silently excluded. The same orientation construction is followed by full-family BY. No finite-sample guarantee was prespecified for this procedure. Bounded Hoeffding–Holm applies only to the bounded IID scenarios. Since $Z \in [-1.1, 1]$, its component p-value is $\exp[-2m\max(\bar{Z}, 0)^2/2.1^2]$ [13], followed by the same orientation and Holm construction. It abstains outside that domain.

The eight original identities are retained unchanged. The ninth method, `legacy_source_screened_BH`, reproduces the actual frozen historical source's arithmetic. Eight `exact_` counterparts of the original methods use explicit rational critical thresholds and exact bootstrap-count grids. Analytic p-values are treated as the exact dyadic values represented by their finite floats, not as infinitely accurate analytic probabilities. Rational harmonic constants define BY thresholds. These additions change numerical semantics, not the sampling assumptions, screening target, or statistical validity of a component test.

A.3 Coverage, availability, power, and Monte Carlo accounting

For each independent study, let R be discoveries and V false discoveries. False-discovery proportion is $V/\max(R, 1)$; its expectation is FDR. Familywise error is $\mathbf{1}\{V > 0\}$, and power is true discoveries relative to the number of alternatives when alternatives exist. Both practical and sign truth are retained. Ordinary interval coverage uses a fixed source 0/policy 0-minus-policy 1 contrast. For historical methods, that interval is generated in a separate unconditional diagnostic stream. Coverage of actually selected intervals is a different, selection-conditional endpoint; it is not replaced by ordinary coverage.

Interval availability is reported separately. In the MCAR and selectively missing scenarios, cluster-t ordinary coverage is based on 1,676 and 202 of 8,000 studies, respectively. Historical selected-interval coverage in the IID equal null is 0.123834 over 958 studies emitting selected intervals, whereas ordinary fixed-contrast coverage is 0.941875 over 8,000 studies. These are study-level averages, not independent interval observations. The complete endpoint CSV also includes null rejection fractions, maximizer-set coverage and size, and deterministic-ID correctness; exact empirical maximizing sets are not confidence sets.

The five possible analysis looks are 500, 1,000, 2,000, 4,000, and 8,000 studies per scenario. For a bounded study-level endpoint with sample variance s^2 and available count N, the two-sided empirical Bernstein half-width is

$$b = \sqrt{\frac{2s^2 \log(4/\delta)}{N}} + \frac{7\log(4/\delta)}{3(N-1)}. \tag{6}$$

The bounds clip $\bar{X} \pm b$ to $[0, 1]$; empty endpoints are unestimated with the full range. The construction uses the empirical Bernstein bound and a finite union over both tails and all reported looks [17]. Conditional coverage uses the count of IID studies emitting an interval, with that availability explicit. The original allocation is $\delta = 0.05/(18 \cdot 8 \cdot 16 \cdot 5)$; the expanded report uses $0.05/(18 \cdot 17 \cdot 16 \cdot 5)$. Dependence among methods and endpoints does not invalidate the union bound. The wider accounting does not erase post-diagnosis method selection.

Stopping uses only the original eight methods. Required half-widths are 0.025 for practical FDR, familywise error, and null rejection fraction; 0.03 for ordinary interval coverage; and 0.08 for defined power. Selected-interval coverage is not a stopping target. The calibration screen flags concern if the practical-FDR lower bound exceeds 0.05 or ordinary-coverage upper bound is below 0.95. Empirical compatibility with tolerance requires FDR upper bound at most 0.06 and coverage lower bound at least 0.94 in relevant assumption-valid scenarios. All other estimable cases are inconclusive. The screen permits conservative overcoverage and is not a proof of exact nominal calibration.

Table 12. Complete original-eight scenario screen and stopping counts. C: calibration concern; I: inconclusive; E: empirical compatibility with tolerance; A: outside-domain abstention. P: original precision objective met; M: maximum-count stop with gaps. H-S/H-F/H-BY: historical screened BH/full BH/plus-one BY; t/c-t/O/G/Hf: IID t/cluster t/oracle normal/grouped bootstrap/bounded Hoeffding. These use the *original-eight* uncertainty family. Full numerical results for all 17 methods and both families are in `stat01-all-endpoints.csv`.

ID	Scenario	Studies	Stop	H-S	H-F	H-BY	t	c-t	O	G	Hf
0	iid_equal_null	8,000	P	C	I	I	I	I	I	C	A
1	iid_stable_null	8,000	P	I	I	I	I	I	I	C	A
2	iid_near_tie_null	8,000	P	C	I	I	I	I	I	C	A
3	exact_ties_null	2,000	P	E	E	E	E	E	E	E	A
4	practical_boundary	8,000	M	C	I	I	I	I	I	C	A
5	small_sign_reversal	8,000	M	C	I	I	I	I	I	C	A
6	medium_reversal	8,000	M	C	C	I	I	I	I	C	A
7	large_reversal	8,000	M	I	I	I	I	I	I	C	A
8	block_null	8,000	M	C	C	C	C	I	I	C	A
9	block_reversal	8,000	M	C	C	C	C	I	I	C	A
10	ar_null	8,000	M	C	C	C	C	C	I	C	A
11	ar_reversal	8,000	M	C	C	C	C	C	I	C	A
12	unequal_variances	8,000	M	I	I	I	I	I	I	C	A
13	mixed_distributions	8,000	M	I	I	I	I	I	I	C	A
14	mcar_reversal	8,000	M	I	I	I	I	I	I	C	A
15	selective_null	8,000	M	C	C	C	C	I	C	C	A
16	bounded_reversal	8,000	P	I	I	I	I	I	I	C	E
17	bounded_reversal_n400	4,000	P	I	I	I	I	I	I	I	E

A.4 Diagnostic outcomes and stopping interpretation

Other scenarios expose the practical-versus-sign target mismatch: practical FDR is 0.39500 at the exact practical boundary and approximately 0.73316 in the medium-reversal scenario. Full scenario results are retained rather than summarized as a single global error rate.

These findings also have limits. Assumption-aligned Gaussian procedures frequently remain *inconclusive* under the stringent simultaneous Monte Carlo screen; that is not evidence that they are miscalibrated. Conservative bounded Hoeffding–Holm comparisons have zero observed false

discoveries and 100% observed diagnostic coverage, but practical power is zero at 40 windows and approximately 0.05263 at 400. The campaign does not certify a generally useful replacement procedure. Of 144 original method/scenario classifications, 45 indicate concern, 74 are inconclusive, 9 meet the empirical tolerance screen, and 16 abstain outside their domain. Seven of the nine compatible entries arise in the degenerate exact-ties scenario; the remaining two are bounded Hoeffding.

Completion and precision must also be separated. Scenarios 3 and 17 stop at 2,000 and 4,000 studies; the other sixteen finish at 8,000. Six scenarios meet the original precision objective, while twelve stop at the maximum count with precision gaps. The original eight-method uncertainty family controls stopping. The expanded 17-method report has its own, wider allocation; additions do not retrospectively become prespecified. Neither maximum-count completion nor an inconclusive calibration screen is a successful calibration claim.

A.5 Numerical boundary case and substantive limitations

The saved scenario 11, replicate 6628, comparison index 47 has a screened family of 43 entries. At rank 43 the stored p-value is 0.05. The original port evaluates $0.05 \times 43/43$ as 0.049999999999999996 and does not reject; the actual source evaluates $(43/43) \times 0.05$ as 0.05 and rejects. The mathematical bootstrap-grid comparison is $1/20 \leq 1/20$ and rejects. The old referee still reproduces its mismatch on this preserved record; the recovery contracts explicitly distinguish all three identities.

The reuse audit found 31 port-versus-source differences among the original 112,275 studies. At final aggregation, the largest difference in endpoint means between those implementations is 0.00025. Only the exact screened-BH addition changes final endpoint means relative to its original counterpart, with maximum change 8.56×10^{-7}; the other seven exact variants' endpoint means match their originals. Aggregate equality does not establish identity of every rejection vector.

These small arithmetic differences do not repair the larger statistical problems. Full-family historical BH has practical FDR 0.51875 in the block null and 0.778625 in the AR null. Grouped-bootstrap BY has ordinary coverage 0.88775 in the IID equal null and 0.759875 in the AR null. Under AR, cluster-t coverage is 0.8600 in the null and 0.864375 in the reversal scenario because fixed five-window clusters remain dependent. Conversely, low FDR can coexist with negligible power: in the block-reversal scenario, cluster-t practical power is approximately 0.00000658, oracle-normal 0.000296, and grouped-bootstrap zero. Bounded Hoeffding power is zero at $n = 40$ and 0.052632 at $n = 400$, with original-family bounds approximately $[0, 0.00401]$ and $[0.04462, 0.06065]$. Inconclusiveness, abstention, and conservatism remain findings rather than reasons to select a new reporting threshold.

B Evidence Identities and Reproduction Scope

The C01 full run is `c01-full-167b9074e5711420`; the recovery run is `stat01-recovery-v2-061933a64e27da20`. Execution commits are `2d8342e699f35044044ac518c0b75a3c4a1c768c` and `1ac4c8dd1421aabad05e6ce243cb911a6ed6c8e1`, respectively. The independent evidence check is pinned at commit `6b2eedd7aa254986b85c1bb901587641cbeac8b0`; the supporting analysis at `d22c7751af9c563d5a4d8a304c007f26112c4d12`. The canonical frozen-window hash begins `0d1aa06ccbee`; the physical window-file hash begins `97fbaf6a4b9b`; absolute C01 metrics begin `5700c4bbd738`. Full hashes, source paths, and stored-copy hashes are supplied in `evidence/input-hashes.json` and the input manifest.

The completed C01 campaign performed exhaustive event verification during execution. The final review separately checked all 5,760 terminal records and recomputed their metrics, all 164 launch-file fingerprints, all 144 reused identities, and all 720 request-condition records. Its fresh event replay covered 24 new-cell journals, structurally chosen from the first nonpilot window per

source at HIGH_PRESSURE, both capacities, and all adaptations. Independent report checking covers 5,760 absolute rows, 15,840 rankings, 95,040 matched differences, and 26,136 reversal groups. The reproduction scripts regenerate displays only from those saved summaries and verifies their fingerprints.

STAT01's recovery validated all 5,360 batches during execution. The final review checked the complete batch inventory and independently recombined all final endpoint moments with high-precision arithmetic; its fresh record-level check covered 200 saved studies across eight scenarios, including the original failing record. It did not claim to rerun all raw studies. Recovery preserves original bytes and labels post-diagnosis comparisons separately. Full raw-record validation and display regeneration are different reproducibility tasks. The manuscript packet supplies the latter, and the separate local candidate contains the complete saved raw records with verification recipes; neither workflow generates new outcomes.

The separate budget sensitivity uses the same pinned C01 engine and referee. Its prospective protocol, exact condition identities, source and runner hashes, per-cell outcomes, and independent analysis are retained in `budget-sensitivity/`. Its run hash begins `f19883ccf2c4`. The 12 pilot cells are reused in the 72-cell total, not independent replicates. A documented JSON resume-metadata correction changes no policy, engine, input or configuration. Full new journals remain in the local sensitivity artifact; no new dataset release is asserted. The subsequent `urgency-analysis/` audit reconstructs these 18 SLO-gated histories and checks the finite-work certificate using separate event-ledger and per-request-interval implementations. It executes no new scientific cells; supplied-state predicate tests are engineering checks.

The historical evidence remains under `paper/tompecs-m2`, including its statistical audit, claim changes, cross-metric set comparisons, and finite-reference ranking-recovery analyses. Those materials remain archived support rather than additional independent experiments. Related headroom and module-intervention projects share some infrastructure and topics, but their causal or native-engine results are not used as evidence for this paper. Historical windows, input reconstruction, and the archived 36/990 rule result are reused explicitly; C01 has separate execution semantics and STAT01 separate synthetic truth. Exact overlap with some older cross-project BurstGPT selections remains unresolved. This uncertainty limits claims of independent evidence across projects, not the recorded within-C01 nonoverlap check.

C Chronological Summaries

Figure 4 reports SLO-minus-chunked ANWG differences in four consecutive ten-window blocks. Across settings and capacities, block means range from −0.0120 to +0.2030 for Azure, −0.0300 to +0.2330 for Bailian, and +0.0150 to +0.3275 for BurstGPT. Negative differences occur in 8/48 Azure and 24/48 Bailian block/setting/capacity combinations, and none of the 48 BurstGPT combinations. The frozen EARLY/MIDDLE/LATE summaries show the same qualitative contrast: negative differences in 6/36, 20/36, and 0/36 combinations, respectively. These are corpus-order sensitivities, not estimates of a temporal process or guarantees for future traffic.

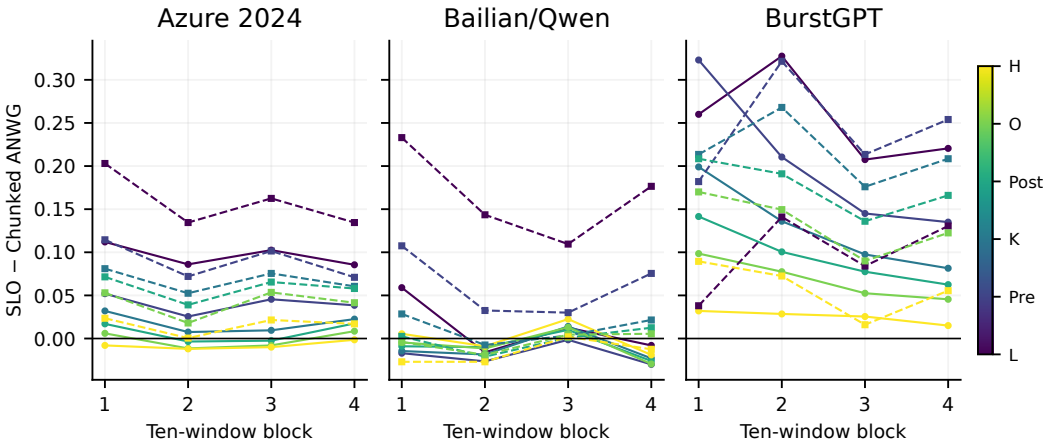


Fig. 4. Chronological block means of SLO-gated minus chunked-FCFS ANWG. Each block contains ten frozen windows. Colors identify inherited settings (L through H); solid circles and dashed squares denote four and eight slots. Blocks have unequal durations and are not independent temporal samples. The figure uses all 144 block/setting/capacity combinations.